

GENERAL THETA FOUNDATIONS I: STATISTICAL RESOURCE DUALITY AND TWO-PREPARATION PHYSICAL CERTIFICATION

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ABSTRACT. We study the resources needed to distinguish a marked experiment from a family of private causal simulators. We separate the optimization of one finite architecture from the optimization over all admissible architectures. For reset experiments whose feedback rows are garblings of a common row, we express the unrestricted-memory preparation value as a minimum of posterior-predictive total variations. The resulting dual bounds every adaptive finite-memory architecture, without selecting an architecture in advance. For the positive-noise marked collision experiment, a four-point rational dual certificate excludes all auditors with at most one training preparation. A deterministic auditor with two training preparations attains a strictly larger uniform score. Its complete bit-level register profile has peak twelve, proving that two training preparations are necessary and sufficient at every peak budget at least twelve on an explicit interval of score thresholds. For general finite architectures we give subdivided positive-basis certificates of explicitly bounded degree and error, including reused stochastic rows. We also give a fully priced autonomous binary-testing construction, with the stationary bound attributed to Hellman and Cover. The earlier compatible-occupation, noisy-testing, continuation and physical-comparison developments are retained in the accompanying complete mathematical volumes.

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1. INTRODUCTION

An exact description of finite stochastic controllers does not by itself determine a resource frontier. There are two distinct questions. The first asks for the best controller within a specified architecture. The second asks whether any architecture satisfying a resource budget can perform a specified statistical task. A lower bound for the first question need not answer the second. We develop certificates that respect this distinction.

The principal application is a marked collision experiment. Its ideal all-on law is supported on four atoms, but neither the simulator nor the auditor is restricted to a four-atom transcript. The simulator produces two independent report bits and an independent actual mark; the target has a correlation between its reports and that mark. The auditor may use the original feedback rows, arbitrary stochastic transitions and stopping rules, subject to the reset and validation interface. Our upper certificate for the auditor is valid simultaneously for all such architectures. The matching construction uses the mark during validation and not during training. This distinction permits a much shorter preparation schedule than an audit based on unmarked diagonal events.

Put

$$(1) \quad u_1 = \frac{4035777}{10240000}, \quad m_2 = \frac{12 - 3\sqrt{3}}{16}, \quad \beta_0 = \frac{1}{300}.$$

The main resource conclusion is

$$(2) \quad N_c^{\text{phys}}(W, Q) = 2 \quad \text{if} \quad W \geq 12, \quad \text{TV}(Q, Q_*) \leq \beta, \quad u_1 + \beta \leq c < m_2 - \beta.$$

Here N counts candidate training preparations; two additional, fresh validation preparations are charged separately. The objective is a strict expected signed score greater than c . In particular, (2) applies to $c = 2/5$ and $\beta = \beta_0$. The lower bound permits unlimited memory, whereas the construction has peak width twelve. This is an exact preparation boundary on a high-width segment, not a claim that twelve is the least possible peak width.

Three mechanisms organize the proof. First, a dominating experimental row removes the outer choice of feedback schedules from the converse. Second, minimax duality turns a prior on fixed simulator parameters into a posterior-predictive upper certificate. Third, residual continuation functions compile the winning decision into a compatible small-memory machine. The general algebraic encoding is useful for checking that the compiled machine belongs to the intended class, but no generic Positivstellensatz is needed to prove (2).

For finite architectures we also supply quantitative information not contained in eventual sum-of-squares completeness. A partition of the stochastic-row simplices into cells of diameter $O(m^{-1})$ gives positive-basis certificates with error at most HS/m , where H bounds the number of row uses and S bounds the number of destinations in a row. The block degree is the maximum number of times the corresponding row can be used. The argument works with autonomous row reuse. It gives an effective, generally exponential, certification procedure; it is not a uniform degree bound for the unmodified Putinar hierarchy.

1.1. Relation to the literature. Comparison of experiments and minimax theory provide the classical background [1, 2]. Polynomial feasible-frequency formulations for partially observed stochastic control are treated by Müller and Montúfar [3]; the rank-one compatibility constraints below are not presented as a new general conditional-independence device. Likewise, strict positivity certificates use established real algebraic geometry [4]. Our quantitative subdivision theorem is a positive-basis construction with a causal error estimate, rather than a new general Positivstellensatz.

Finite-state testing has a substantial earlier theory. Hellman and Cover [5, 6], Flower and Hellman [7], and Cover, Freedman and Hellman [8] distinguish time-invariant rules, finite horizons

and randomization. We explicitly identify the stationary result in Section 6 as a specialization of that theory. Managoli and Prabhakaran [9], posted in May 2026, study randomized time-invariant finite-state binary testing against adaptive adversarial distribution choices and establish matching state exponents, with exact bounds for a class of problems. Our physical theorem has a different objective: a finite training–validation signed score against a fixed private simulator, with bit-level memory and preparations charged. We make no priority claim for the general phenomenon that randomization can save states.

Norberg’s filtered comparison of experiments [10] is pertinent to the broader causal comparison program. The original proof-level comparison remains incomplete: the accessible publisher record verifies the bibliographic identity but does not justify an assertion that its results exclude the present constructions. None of our proofs uses such an exclusion.

1.2. The mathematical volumes. The present article gives the new proof chain. The accompanying *Complete Mathematical Manuscript* contains this article followed by the unaltered 96-page revision 23. The *Complete Development* contains this article followed by the unaltered 649-page development. Thus the earlier general continuation, occupation, noisy binary, physical, Gaussian and operator interfaces remain available with their full hypotheses and proofs [11]. They are not silently replaced by (2). In particular, the independent A2 geometric chain and the unresolved historical B4 and broad C2 obligations remain distinct. The mathematical role of the present first foundation is an experimental and resource-comparison root; a universal derivation of every program branch is not assumed.

2. EXPERIMENTS, ARCHITECTURES AND ATTAINABLE RESOURCES

Throughout, total variation is $\text{TV}(P, Q) = \sup_{0 \leq h \leq 1} (P - Q)h = \frac{1}{2} \sum_x |P(x) - Q(x)|$. An experimental parameter is unknown and fixed throughout an execution. Controller randomization is independent of that parameter. A reset preparation is a fresh draw, independent of earlier preparations given the parameter. A controller may not recover discarded information from an uncharged tape, clock, selector or calibration table.

Definition 2.1 (Finite architecture). A finite architecture A specifies a finite causal execution graph, its legal experimental operations, the observable inputs to each controller row, and a register alphabet at every cut. It also specifies which row occurrences share a stochastic table. A clocked architecture may use different rows at different scheduled times. An autonomous architecture may use time only through a charged state. For a parameter family Θ and terminal reward in $[-1, 1]$, let

$$V(A) = \sup_{q \in \prod_r \Delta_{sr}} \inf_{\theta \in \Theta} g_\theta(A, q).$$

If $\mathcal{A}(R)$ is the class of architectures obeying a global resource budget R , its outer value is

$$V^{\text{out}}(R) = \sup_{A \in \mathcal{A}(R)} V(A).$$

Both the architecture and the simulator parameter are fixed before the controller’s random choices. In particular, a mixture of complete controllers is not automatically an admissible controller at the original register profile.

For a finite hidden execution graph, compatibility can be stated without division. Let $x_{\theta ir}$ be the incoming mass of hidden history i at observable row r , and let $f_{\theta irj}$ be the outgoing mass through choice j . Include the usual fixed-kernel flow equations, as well as

$$(3) \quad f_{\theta irj} \geq 0, \quad \sum_j f_{\theta irj} = x_{\theta ir}, \quad f_{\theta irj} x_{\theta' i' r} = f_{\theta' i' rj} x_{\theta ir}.$$

For shared rows, the indices i, i' include all their occurrences.

Proposition 2.2 (Compatible realization). *For a specified finite architecture, nonnegative flow arrays satisfying (3) are exactly those realized by a common stochastic row at every declared observable input, including at zero incoming mass.*

Proof. Necessity follows from $f_{\theta irj} = x_{\theta ir} q_r(j)$. For sufficiency, if some incoming mass at row r is positive, define $q_r(j)$ by dividing the corresponding outgoing mass by that incoming mass. Conservation makes q_r stochastic, and the cross-products force the same ratio at every other positive incoming mass. If an incoming mass vanishes, nonnegativity and conservation force all its outgoing masses to vanish. If the entire row is unreachable, choose any legal stochastic row. Induction through the finite execution graph reconstructs the prescribed flows. The same ratio argument applies to identified occurrences of an autonomous row. $\square$

This formulation is inherited from the complete companion and is included to fix the controller class. Compact semialgebraic feasibility and an Archimedean positivity theorem provide a complete semialgebraic certificate hierarchy for a *fixed finite architecture*. They do not interchange the two optimizations in the definition. Section 5 gives a quantitative alternative for the inner problem. The next section supplies a different certificate that survives the outer optimization.

3. A STATISTICAL DUAL THAT IS UNIFORM OVER ARCHITECTURES

Let Θ be compact, let Ω be finite, and let $\theta \mapsto P_\theta \in \mathcal{P}(\Omega)$ be continuous. Fix $Q \in \mathcal{P}(\Omega)$. An N -preparation reset selector observes $Z = (Z_1, \dots, Z_N) \sim P_\theta^{\otimes N}$ and chooses a response $h_Z \in [0, 1]^\Omega$. Its expected validation score is

$$(4) \quad G_\theta(h) = \sum_{z \in \Omega^N} P_\theta^{\otimes N}(z) (Q - P_\theta) h_z.$$

The response is fixed before two independent validation draws. Operationally it can be realized by choosing a random event with those coordinate marginals and retaining that event for both validations. No memory bound is imposed in the definition of $U_N(P, Q) = \max_h \min_\theta G_\theta(h)$.

Theorem 3.1 (Posterior-predictive dual). *Both extrema in the following identity are attained:*

$$(5) \quad U_N(P, Q) = \min_{\mu \in \mathcal{P}(\Theta)} \sum_{z, \omega} \left[\int P_\theta^{\otimes N}(z) \{Q(\omega) - P_\theta(\omega)\} d\mu(\theta) \right]_+$$

$$(6) \quad = \min_{\mu \in \mathcal{P}(\Theta)} \sum_z m_\mu(z) \text{TV}(Q, \bar{P}_\mu(\cdot | z)).$$

Here $m_\mu(z) = \int P_\theta^{\otimes N}(z) d\mu(\theta)$ and, when $m_\mu(z) > 0$,

$$\bar{P}_\mu(\omega | z) = \frac{\int P_\theta^{\otimes N}(z) P_\theta(\omega) d\mu(\theta)}{m_\mu(z)}.$$

A zero-mass term in (6) is defined to be zero.

Proof. The response space is a finite-dimensional compact convex box. Probability measures on the compact space Θ form a weakly compact convex set. The integral of (4) is continuous and affine in each argument. The minimum over parameters equals the minimum over probability measures. The minimax theorem [2] therefore interchanges the maximum and minimum. For fixed μ , each response coordinate is independent in the maximization and its contribution is the positive part of its coefficient, proving (5). The coefficient vector at a fixed z has zero sum. For positive $m_\mu(z)$ it equals $m_\mu(z)(Q - \bar{P}_\mu(\cdot | z))$, proving (6). If $m_\mu(z) = 0$, every nonnegative likelihood integrand vanishes μ -almost surely and all coefficients are zero. Continuity on the two compact strategy spaces gives attainment. $\square$

Theorem 3.2 (Dominating-row converse). *Suppose every permitted training row is a parameter-independent causal garbling of one fresh draw from P_θ . Suppose every permitted validation row is the same garbling of the target and candidate laws, and every selected row–event pair is fixed before validation. Then $U_N(P, Q)$ bounds the worst-case expected score of every auditor using at most N candidate training preparations, regardless of its register profile, feedback schedule, independent randomization, or stopping rule. Independent target-only calibration, whose law does not depend on θ , does not invalidate the bound. If the all-on row and all its events are permitted, the bound is attained in the class with no memory cap and N training preparations.*

Proof. Generate N independent all-on observations in advance. At the t th requested preparation, apply the declared garbling to observation t , using only previously simulated reports and independent randomization. This simulates the adaptive transcript with its exact law. A stopped execution ignores the unused observations; it does not reveal them to the auditor. Pull back its final row–event response to the all-on alphabet. Conditional on the entire generated training word and the independent random seed, this is a function in $[0, 1]^\Omega$. Averaging the seed, including any target-only calibration, produces one response h_z for each full word z . The score is consequently (4). This proves the upper bound for the union of all architectures. Conversely, retaining the entire word and then a selected event implements any such finite response kernel when memory is unrestricted. Fresh validation gives exactly (4). $\square$

Remark 3.3 (What the dual does not convexify). The measure μ is an adversarial prior used to bound the minimum over *fixed* parameters. Its likelihood is $\int P_\theta^{\otimes N} d\mu(\theta)$, not $(\int P_\theta d\mu(\theta))^{\otimes N}$. It neither enlarges the private candidate class to mixtures nor gives the auditor a free persistent selector. Attainment of U_N without a memory cap does not assert attainment at any specified finite profile.

The state in (6) is generated by the actual preparation law, not by a geometric measure chosen independently of the experiment. For sequential continuation one retains the posterior μ_z or its future-test equivalence class; a single predictive marginal need not be closed under further observations. This is the distinction between a one-step response quotient and a future-predictive quotient.

Corollary 3.4 (Finite moment support for the product family). *For two independent Bernoulli report bits with parameters (p, q) and an independent fair mark, an optimal prior in (5) can be chosen with at most $(N + 2)^2$ atoms. The two report counts (k, ℓ) are a causally updatable sufficient statistic for computing the dual.*

Proof. Conditional on the two counts, all arrangements of report bits and all mark words have parameter-independent probabilities. Summing their coefficients reduces (5) to the $(N + 1)^2$ count pairs. Their likelihoods are

$$\binom{N}{k} p^k (1 - p)^{N-k} \binom{N}{\ell} q^\ell (1 - q)^{N-\ell}.$$

After multiplication by $Q(\omega) - P_{p,q}(\omega)$, every integrand has degree at most $N + 1$ in each variable. Thus the objective depends only on the $(N + 2)^2$ rectangular moments. Their constant coordinate is one. Carathéodory’s theorem in the remaining affine space replaces an optimal moment vector by a convex combination of at most $(N + 2)^2$ point evaluations. Counts update by addition of the current bits. No claim that this count register is optimal for the decision is made. $\square$

4. THE PHYSICAL EXPERIMENT AND ITS SHARP PREPARATION BOUNDARY

Take $\Omega = \{0, 1\}^3$, with coordinates (x_1, x_2, w) . The all-on private candidate is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad 0 \leq p, q \leq 1.$$

The mark w is the actual mark, not an invented simulator bit. Put $a = (1 - p)(1 - q)$ and $c = pq$. On the ordered atoms

$$(7) \quad 000, \quad 001, \quad 110, \quad 111,$$

let Q_* have masses $5/16, 3/16, 3/16, 5/16$, and set it to zero elsewhere. The candidate masses on these atoms are $a/2, a/2, c/2, c/2$. A target Q is allowed whenever $\text{TV}(Q, Q_*) \leq \beta$.

The original feedback operation has the form

$$(8) \quad T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w), \quad g : \{0, 1\} \rightarrow \{0, 1\}.$$

The selected gate is observable. It may depend on earlier preparations and on the first current report. The target rows are pushforwards by (8). For the converse, restrict the adversary to the admissible private subfamily whose second emission is AX_2 , where A is the current gate and $X_2 \sim \text{Ber}(q)$ is independent of the first emission and of the fair mark. Every row is then a causal garbling of $b(p, q)$. For the construction only the all-on row is used, and every private candidate in the original width-one class has the all-on product law just specified. These two facts, proved for the microscopic interface in [11, physical-complexity, feedback-completion lemma], explain why using a subfamily for the lower bound does not narrow the upper construction.

The admissible audit first trains, then selects one row–event pair, then uses one fresh candidate and one fresh target validation. It reports $D = \mathbf{1}_A(Y) - \mathbf{1}_A(X)$. The event and row cannot be reselected after seeing a validation draw. All bit, mark, decision and validation cuts are charged. Define $N_c^{\text{phys}}(W, Q)$ to be the minimum number of candidate training preparations among these original-interface auditors, with peak register width at most W , whose score is uniformly strictly greater than c . Randomization, feedback and stopping are allowed. If no such audit exists, the value is $+\infty$.

Theorem 4.1 (Two-preparation physical segment). *Let u_1, m_2 be as in (1). For every $\beta \geq 0$, $\text{TV}(Q, Q_*) \leq \beta$, and*

$$W \geq 12, \quad u_1 + \beta \leq c < m_2 - \beta,$$

one has $N_c^{\text{phys}}(W, Q) = 2$. The two-preparation audit is deterministic, uses no acquired calibration and no random bits, and has the full profile

$$(9) \quad (1, 2, 3, 3, 4, 5, 5, 10, 12, 10, 10, 7, 3)$$

from the initial cut through its twelve bit/mark updates. In particular, its decision alphabet has five labels and its peak has twelve. The lower bound remains valid with no memory cap.

The proof occupies the next three subsections. The interval in the statement can be empty for large β ; no assertion is then made. For $\beta = 1/300$, it contains $c = 2/5$ strictly between its endpoints.

4.1. An exact rational certificate against one preparation.

Lemma 4.2. *Every original-interface auditor using at most one candidate training preparation has worst-case score at most $u_1 + \beta$, regardless of memory.*

Proof. By Theorem 3.2, it suffices to bound $U_1(b, Q_*)$ and then perturb the target. In (5) use the four fixed parameters

$$(p, q) = (1/8, 1/8), (1/5, 1/5), (4/5, 4/5), (7/8, 7/8)$$

with weights $1/16, 7/16, 7/16, 1/16$, respectively. This measure is not asserted to be least favorable. Since $p = q$ on its support, the training count $K = x_1 + x_2$ is sufficient. The independent training mark and the order of unequal bits contain no further information about this prior.

For $k = 0, 1, 2$ and the four validation atoms in (7), the signed coefficients in (5), after grouping by K , are the entries of

$$(10) \quad \frac{1}{40960000} \begin{pmatrix} -1977 & -1775177 & 2170423 & 3943623 \\ 1765554 & 191954 & 191954 & 1765554 \\ 3943623 & 2170423 & -1775177 & -1977 \end{pmatrix}.$$

For example, the coefficient in column j is

$$\sum_{i=1}^4 \mu_i \binom{2}{k} p_i^k (1-p_i)^{2-k} \left(Q_{*,j} - \frac{(1-p_i)^2 \mathbf{1}_{j \leq 2} + p_i^2 \mathbf{1}_{j \geq 3}}{2} \right).$$

This identity proves every entry of (10) by rational arithmetic. All coefficients at off-diagonal validation atoms are nonpositive. The sum of the positive entries in (10) is $4035777/10240000 = u_1$. Each response coordinate is in $[0, 1]$, so this sum bounds its prior-averaged score and hence its minimum over the four fixed candidates.

Replacing Q_* by Q changes the score of any trained response by at most β . The training law is a candidate law and is unchanged. Consequently the same prior bounds every score by $u_1 + \beta$. A procedure with no training, or one that stops before taking its permitted sample, is covered by ignoring the generated all-on observation. $\square$

4.2. A marked five-event selector. Take two independent all-on training preparations. Let S be the total number of ones among their four report bits; discard their marks. Choose the following events, represented by their indicators on (7) and with indicator zero at every off-diagonal atom:

$$(11) \quad \begin{array}{c|cccc} S & 000 & 001 & 110 & 111 \\ \hline 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & 0 \end{array}$$

Denote these indicators by $h_0, \dots, h_4$ and report $D = h_S(Y) - h_S(X)$ on independent validation draws.

Lemma 4.3 (Exact uniform score of the selector). *Against the ideal target, the selector (11) has uniform minimum score $m_2 = (12 - 3\sqrt{3})/16$. Against Q , its score is at least $m_2 - \beta$.*

Proof. The generating polynomial of S is

$$((1-p) + pz)^2((1-q) + qz)^2.$$

Writing $g_s = (Q_* - b(p, q))h_s$, its expected score is $G(p, q) = \sum_{s=0}^4 [z^s]((1-p) + pz)^2((1-q) + qz)^2 g_s$. Substitution of the five rows in (11) and collection of terms with $x = p + q - 1$, $y = p - q$, $u = x^2$ and $v = y^2$ gives

$$(12) \quad \begin{aligned} G(p, q) = H(u, v) = & -\frac{u^3}{32} + \frac{3u^2v}{32} + \frac{3u^2}{16} - \frac{3uv^2}{32} - \frac{uv}{4} - \frac{3u}{32} \\ & + \frac{v^3}{32} + \frac{v^2}{16} + \frac{15v}{32} + \frac{7}{16}. \end{aligned}$$

This is a polynomial identity, not an interpolation over a grid. Since $0 \leq u \leq 1$ and $v \geq 0$ on the parameter square,

$$\partial_v H(u, v) = \frac{3(u-v)^2 - 8u + 4v + 15}{32} \geq \frac{7}{32}.$$

Thus $H(u, v) \geq H(u, 0) = (-u^3 + 6u^2 - 3u + 14)/32$. The derivative of the last expression is $(-3u^2 + 12u - 3)/32$. On $[0, 1]$ it changes sign from negative to positive at $u = 2 - \sqrt{3}$. Substitution gives $(12 - 3\sqrt{3})/16$. The minimizing values are feasible: take $p = q = (1 \pm \sqrt{2 - \sqrt{3}})/2$. This proves exactness of the minimum for this specified selector, not optimality of its score over all selectors with two preparations. Finally, for each s , the target expectation of $h_s \in [0, 1]^\Omega$ changes by at most β . Averaging over training proves the claimed robustness. $\square$

4.3. Causal compilation and the full register profile.

Lemma 4.4. *The selector and its two validations admit the deterministic profile (9). No selector, first-bit buffer or validation accumulator is outside the charged register. An autonomous phase-tagged realization uses at most 75 states.*

Proof. After each training report bit retain the count accumulated so far. The two marks are read and discarded. The first six updates therefore have widths 2, 3, 3, 4, 5, 5 after the initial singleton. At the decision cut the five count values select the five events.

During candidate validation, after the first bit retain the event and that bit, using at most ten states. After the second bit retain the event and the function of the unread mark that gives its candidate-event indicator. For events 0, 1, 2, 3, 4, respectively, the possible functions are

$$\{0, 1\}, \quad \{0, 1 - w, 1\}, \quad \{0, 1\}, \quad \{0, w, 1\}, \quad \{0, 1\}.$$

There are twelve event–function pairs. After the candidate mark, retain the event and its now known candidate indicator $c_0 \in \{0, 1\}$, using at most ten states.

During target validation, after the first bit discard the event name and retain c_0 and its residual function of the second bit and mark. These residual functions belong to

$$\{0, x_2, (1 - x_2)(1 - w), 1 - x_2, x_2 w\}.$$

This uses at most ten states. After the second bit, the residual *difference* as a function of the unread mark belongs to

$$\{-1, 0, 1, w, 1 - w, w - 1, -w\},$$

so seven states suffice. The mark update gives the three possible outputs $-1, 0, 1$. Each transition depends only on its retained residual and the current report. Thus it is a legal common observable row, not a family of independently optimized hidden-history maps.

For completeness, equality of two residual functions implies equality of their restrictions after either next bit, so identifying equal residuals is closed under every subsequent update. The supplied transition tables are obtained by these restrictions and can be reconstructed directly from (11). Enumerating residuals is a convenience for reproducing the tables, not the proof of their sufficiency. Finally, disjointly tag the states at the thirteen cuts by their phase and use terminal self-loops. The sum of the displayed widths is 75. This implementation charges the clock; it does not assert that 75 is optimal. $\square$

Proof of Theorem 4.1. Lemma 4.2 excludes $N \leq 1$ for a strict score greater than $c \geq u_1 + \beta$, for every architecture and every memory budget. Lemmas 4.3 and 4.4 supply a two-preparation architecture of peak twelve with score at least $m_2 - \beta > c$. Larger width budgets can use the same machine. $\square$

Corollary 4.5 (The positive-noise collision family). *For the original marked collision targets with $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$, the inherited microscopic estimate $\text{TV}(Q_{d,\sigma}, Q_*) < 1/300$ implies*

$$N_{2/5}^{\text{phys}}(W, Q_{d,\sigma}) = 2 \quad (W \geq 12).$$

There are four preparations in the complete audit: two candidate training, one candidate validation, and one target validation.

Proof. Apply Theorem 4.1. The lower-end margin is the exact rational number

$$\frac{2}{5} - u_1 - \frac{1}{300} = \frac{78269}{30720000} > 0.$$

For the other endpoint, $\sqrt{3} < 7/4$ implies $m_2 > 27/64$ and $27/64 - 1/300 > 2/5$. The physical total-variation estimate, including the actual mark and positive sensor noise, is proved in the retained microscopic-composition and physical-complexity modules [11]; the current deduction does not replace those hypotheses by a zero-noise limit. $\square$

Remark 4.6 (The part of the frontier determined here). The theorem evaluates the minimum preparation coordinate for $W \geq 12$. It does not determine $N_c^{\text{phys}}(W, Q)$ for $3 \leq W \leq 11$, the maximum score with two preparations, or the least peak width at $N = 2$. The earlier exact decision-width result $\kappa_{2/5}^{\text{rows}} = 3$ is unchanged: it allows longer training, whereas this shortest audit uses five decision states. The older points (399, 40602), (320000, 1604) and $(12800 \cdot 2^{400}, 12)$ remain valid constructions with decision width three. They are dominated by (2, 12) in the projection onto training preparations and peak width, but not necessarily in a resource order that also constrains the decision alphabet separately.

5. QUANTITATIVE CERTIFICATES FOR A FINITE COMPATIBLE ARCHITECTURE

We return to an arbitrary finite architecture, now with a finite parameter set. It can have hidden states, controlled experiments, more than two hypotheses and registers of any size. Enumerate its distinct stochastic rows by $r = 1, \dots, R$ and write $q_r \in \Delta_{s_r}$. A reused autonomous row appears only once in this product. Assume each execution uses at most H controller rows, and uses row r at most d_r times. Delete unused rows and put $S = \max_r s_r$. Fixed environmental kernels are not controller rows.

For an integer $m \geq 1$, intersect Δ_{s_r} with the coordinate boxes of side $1/m$. The resulting nonempty closed polytopes cover the simplex, with overlaps permitted at their boundaries. Write C for a product of such row cells and choose a vertex q_r^C in each factor. A row-cell vertex has at most one coordinate not fixed at a box boundary; consequently it has at most $v_r = s_r 2^{s_r - 1}$ vertices, all on the $1/m$ grid. There are at most $m^{\sum_r s_r}$ product cells.

Write the vertices of row cell C_r as $v_{r,j}$ and parametrize $q_r = \sum_j \lambda_{r,j} v_{r,j}$ with λ_r in a simplex. A reward polynomial has degree at most d_r in block λ_r . Use the homogeneous simplex Bernstein basis of degree d_r in that block, obtaining

$$(13) \quad g_\theta(q(\lambda)) = \sum_\alpha b_{\theta,C,\alpha} B_\alpha(\lambda), \quad B_\alpha \geq 0, \quad \sum_\alpha B_\alpha = 1.$$

Degree elevation includes the factors $\sum_j \lambda_{r,j} = 1$. No independence between different visits of a shared row is imposed on its unknown row vector.

Lemma 5.1 (Causal interpretation of the coefficients). *Every coefficient in (13) satisfies*

$$|b_{\theta,C,\alpha} - g_\theta(q^C)| \leq HS/m.$$

Proof. For each row r , independently prepare an ordered list of d_r vertex labels, sampled independently with law λ_r . At its k th visit, use the stochastic row given by the k th vertex label in that list. Before conditioning, the next label is independent of the previous execution, so this experiment has exactly the row q_r at every visit and expected reward $g_\theta(q(\lambda))$. Unused labels may be sampled after termination without altering the reward.

Condition on the counts of every vertex label in every list. The probability of those counts is $B_\alpha(\lambda)$. Conditional on the counts, each list is a uniformly ordered multiset. Its expected reward is independent of λ and is therefore the corresponding Bernstein coefficient. This also proves (13) directly.

Each row used by this conditioned experiment is a vertex of C_r or a mixture of its remaining vertices. Every two vectors in C_r have coordinate differences at most $1/m$, hence total variation at most $s_r/(2m) \leq S/(2m)$. Couple the conditioned experiment to the one using q^C , using the same environmental randomness while their executions agree. At each actual controller update the probability of a first row-level disagreement is at most $S/(2m)$. There are at most H such updates on the agreeing path. The terminal laws consequently differ in total variation by at most $HS/(2m)$. A reward in $[-1, 1]$ has expectation difference at most twice that quantity. This proves the estimate. The conditioned urn is used to analyze a coefficient; it is not declared to be an admissible free-memory controller. $\square$

Theorem 5.2 (Explicit subdivision certificate). *Define*

$$\ell_C = \min_{\theta} g_{\theta}(q^C), \quad u_C = \min_{\theta} \max_{\alpha} b_{\theta, C, \alpha}, \quad L_m = \max_C \ell_C, \quad U_m = \max_C u_C.$$

For the exact all-stochastic value of the specified architecture,

$$(14) \quad L_m \leq V(A) \leq U_m \leq L_m + HS/m.$$

On each cell, a witnessing parameter θ_C supplies the nonnegative coefficient certificate

$$u_C - g_{\theta_C}(q(\lambda)) = \sum_{\alpha} (u_C - b_{\theta_C, C, \alpha}) B_{\alpha}(\lambda).$$

Its degree in row block r is d_r , and its total degree is at most $\sum_r d_r$. At most $\prod_r \binom{d_r + v_r - 1}{v_r - 1}$ coefficients per parameter and cell are needed. If the primitive data are rational, all coefficients and bounds can be checked by rational arithmetic. Taking $m \geq HS/\varepsilon$ gives a certified interval of length at most ε .

Proof. The point q^C is feasible, so ℓ_C is a lower bound on the value restricted to C . For any $q \in C$, choose a convex representation of its rows. Equation (13) implies

$$\min_{\theta} g_{\theta}(q) \leq \min_{\theta} \max_{\alpha} b_{\theta, C, \alpha} = u_C.$$

Choose a parameter attaining the minimum defining ℓ_C and apply Lemma 5.1; this gives $u_C \leq \ell_C + HS/m$. Taking the maximum over the finite cover proves (14). Choose θ_C attaining the definition of u_C ; all displayed coefficients of its certificate are nonnegative. The degree and count statements follow from the product Bernstein basis. Cell vertices are rational grid points, and a finite execution sums finite products, proving the rational statement. $\square$

Remark 5.3. The number of cells and coefficients can be enormous. Neither polynomial running time nor a uniform degree bound for the original sum-of-squares hierarchy follows. Refinement changes the domain cover, not merely a single global SOS degree. For a finite collection of architectures obeying a specified budget, maxima of these intervals give outer intervals; an unbounded architecture class needs a separate uniform argument such as Theorem 3.2. For compact parameter families, an effective net and a verified modulus for the primitive kernels are additional inputs, not consequences of compactness alone.

The resource estimate also makes row precision explicit. With $m = 2^b$, the lower witnesses have dyadic rows and lose at most $HS2^{-b}$ in value. Thus increasing row precision provides a quantified approximation for a fixed architecture, while its random-bit implementation still has to be charged. The physical theorem avoids this cost altogether because its rows are deterministic.

6. AUTONOMOUS NOISY TESTING WITH A PRECISION AND TIME LEDGER

This section supplies a substantive autonomous comparison rather than merely identifying shared rows. The stationary optimum is classical [5]; the proof and explicit finite-time implementation are recorded to make the resource conventions unambiguous. They are not used as novelty evidence for the physical theorem.

Let reports be independent, with law $\text{Ber}(p)$ under hypothesis $-$ and $\text{Ber}(q)$ under hypothesis $+$, where $0 < p < 1/2$, $q = 1 - p$, and $r = q/p$. A W -state autonomous machine has two time-independent stochastic transition matrices T_0, T_1 . Assume their induced chains $P_- = qT_0 + pT_1$ and $P_+ = pT_0 + qT_1$ are irreducible. A final decision uses only the state, with optional fresh terminal randomization.

Lemma 6.1 (Likelihood-range bound). *If equivalent probability laws ν_+, ν_- on a finite set satisfy*

$$\frac{\max_i \nu_+(i)/\nu_-(i)}{\min_i \nu_+(i)/\nu_-(i)} \leq R,$$

then $\text{TV}(\nu_+, \nu_-) \leq (\sqrt{R} - 1)/(\sqrt{R} + 1)$.

Proof. Write $L = d\nu_+/d\nu_-$, $a = \min L$ and $b = \max L$. Since $\mathbb{E}_{\nu_-} L = 1$, one has $a \leq 1 \leq b$. The chord bound for $(L - 1)_+$ gives

$$\text{TV}(\nu_+, \nu_-) = \mathbb{E}_{\nu_-} (L - 1)_+ \leq \frac{(1 - a)(b - 1)}{b - a}.$$

For fixed b/a , maximizing the last expression gives $a = 1/\sqrt{b/a}$; substitution yields $(\sqrt{b/a} - 1)/(\sqrt{b/a} + 1)$. It is increasing in b/a , proving the assertion. The identical-law case follows directly. $\square$

Theorem 6.2 (Autonomous stationary bound and an explicit approximation). *Put $D = W - 1$, $W \geq 2$. In the irreducible autonomous class the infimum stationary minimax error is*

$$(15) \quad e_W = \frac{1}{1 + r^D}.$$

For every $\epsilon = 2^{-b}$, $b \geq 0$, there is a machine whose rows have denominator dividing 2^{b+1} , and whose minimax error at report time n from any initial state is at most

$$(16) \quad e_W + \epsilon(W - 2) + \exp \left\{ - \left(\frac{p\epsilon}{2} \right)^D \left\lfloor \frac{n}{D} \right\rfloor \right\}.$$

The stationary lower bound is not a finite-horizon lower bound.

Proof. Every positive induced transition has its likelihood ratio in $[r^{-1}, r]$, and the supports of the two induced matrices are identical. The Markov chain tree formula writes each stationary weight as a sum of products over directed spanning trees rooted at that state. Each product has D transitions; its ratio lies in $[r^{-D}, r^D]$. The ratio of the two unnormalized stationary weights lies in the same interval. Normalizing multiplies all state likelihood ratios by a common number, so their maximum-to-minimum ratio is at most r^{2D} . Lemma 6.1 therefore bounds stationary total variation by $(r^D - 1)/(r^D + 1)$. The equal-prior Bayes error is at least $1/(1 + r^D)$,

and a minimax error cannot be smaller than that Bayes error. This is the equal-prior binary specialization of the Hellman–Cover bound.

For the upper approximation, take states $i = 0, \dots, D$, weights $c_0 = c_D = 1$, and $c_i = \epsilon$ at interior states. On report 1, move from i to $i + 1$ with probability $a_i = \frac{1}{2} \min(1, c_{i+1}/c_i)$, if that neighbor exists; otherwise stay. On report 0, move to $i - 1$ with probability $b_i = \frac{1}{2} \min(1, c_{i-1}/c_i)$, if it exists; otherwise stay. All other probability is assigned to staying. Detailed balance gives

$$\pi_+(i) = \frac{c_i r^i}{\sum_j c_j r^j}, \quad \pi_-(i) = \frac{c_i r^{-i}}{\sum_j c_j r^{-j}}.$$

Reflection $i \mapsto D - i$ exchanges these laws. Decide $+$ for $i > D/2$, $-$ for $i < D/2$, and use a fair terminal decision at $i = D/2$ when this state exists. The two errors agree. Conditional on being at an endpoint, the error is exactly e_W . The total interior stationary mass is at most $\epsilon(W - 2)$, since each $r^i \leq r^D$ and the denominator is at least r^D under $+$, with the reflected argument under $-$. This proves the first two terms of (16).

Every existing neighbor transition under either hypothesis has probability at least $p\epsilon/2$, and every self-loop has probability at least $1/2$. From any state, move down to zero and then stay until D steps have elapsed. This yields the common minorization $P_\pm^D(i, \{0\}) \geq \delta = (p\epsilon/2)^D$. Subtracting that common atom gives contraction in total variation by $1 - \delta$ every D steps. Comparison with the stationary law yields $\text{TV}(P_\pm^n(i, \cdot), \pi_\pm) \leq (1 - \delta)^{\lfloor n/D \rfloor} \leq e^{-\delta \lfloor n/D \rfloor}$. A bounded decision changes error by at most this amount, proving (16). Finally, letting n tend to infinity and then ϵ to zero proves that the stationary infimum equals (15). All transition probabilities belong to $\{0, 1, 1/2, \epsilon/2, 1 - 1/2, 1 - \epsilon/2\}$. $\square$

The W states in this theorem refer to an atomic stochastic update after each report. A sequential fair-bit implementation must also retain the report, the current state, a bit counter and a flag. Each nontrivial move probability is $1/2$ or $2^{-(b+1)}$; test for a string of zero bits of the required length, and pad to $b + 1$ bits if a uniform report cycle is required. At most $4W(b + 2)$ states suffice for this implementation, including its bit-phase counter and a ready phase. This conservative bound is an implementation claim, not an optimal precision–state tradeoff. The time index n in (16) is an observation time at which an external analyst reads the autonomous machine, not a free input clock used by its transitions. The adversarial model of [9] is not covered by the independent-report proof above.

7. PHYSICAL CONSUMERS AND THE FIRST-PRINCIPLES CONNECTION

The physical preparation theorem needs no estimate of the target law as a stored input: the five events are fixed and their uniform validity uses a proved neighborhood of Q_* . This removes calibration memory from this particular consumer without asserting that target acquisition is unnecessary in an arbitrary comparison problem. The general acquired calibration theorem, with its full counter register, remains in the complete companion.

Corollary 7.1 (A raw-interface confidence construction). *Let $m = m_2 - \beta > 0$ and $0 < c < m$. Repeat the full four-preparation audit independently R times and write $\bar{D}_R$ for the mean score. Under every private candidate,*

$$\mathbb{P}(\bar{D}_R \leq c) \leq \exp\{-R(m - c)^2/2\}.$$

Under the null in which the independent candidate and target validation laws agree, with the same event chosen before both validations,

$$\mathbb{P}(\bar{D}_R > c) \leq \exp\{-Rc^2/2\}.$$

A clocked implementation uses $4R$ preparations and at most $12(2R + 1)$ persistent states, with no calibration and no random bits.

Proof. The independent block scores lie in $[-1, 1]$ and have mean at least m under a private candidate. Hoeffding's inequality gives the first bound. Under the null, conditional on the selected event, the two independent validation indicators have the same mean. Each block therefore has mean zero, and the same inequality gives the second bound. Retain the integer sum in $\{-R, \dots, R\}$ along with the current twelve-state block. At a block boundary add its terminal score and reset its register. The deterministic external schedule is part of the clocked convention. $\square$

This is an upper construction on the raw marked interface. Its proof does not transfer a lower bound from a deliberately reduced Bernoulli interface, and no optimal confidence-memory frontier is claimed.

Proposition 7.2 (Target transport without a training multiplier). *For the two-preparation audit, changing only the target from Q to Q' changes its expected score by at most $\text{TV}(Q, Q')$. Its candidate training law, register profile and event selector remain unchanged.*

Proof. Condition on the candidate training word. The expectation difference is $(Q - Q')h_S$, whose absolute value is at most $\text{TV}(Q, Q')$. Average over that unchanged training law. $\square$

In particular, every common-preparation transport estimate verified for the original positive-noise collision family can be inserted here without multiplying it by a long training horizon. If candidate kernels also change, their training and validation defects must be added by a causal coupling; the target-only statement does not remove those terms.

The first-principles route is now concrete. The positive experimental kernels and actual reset preparations give the likelihoods in (5); these generate posterior continuation states. The validation-test quotient gives the positive parts in the statistical dual. A prior then supplies an architecture-uniform impossibility certificate. On the constructive side, the five-event decision is compiled by taking its future residual functions, which are closed under observation shifts. These are compatible causal states and determine the charged profile. Thus the preparation law, predictive continuation and finite-resource realization are used in the same upper/lower theorem. This is not a deduction of geometry or nonlinear operator compactness from the existence of finite stochastic rows.

8. DEPENDENCIES, RETAINED RESULTS AND REMAINING DISTINCTIONS

The exact three-report binary noisy frontier, arbitrary-horizon binary-register minimax recursion and compatible Bayesian interval recursion remain as proved in the complete companion. Its exact odd-horizon majority realization is also retained; it is not a new result of this revision. Section 5 applies to registers of any finite width, controlled rows and any finite parameter set, but provides certified intervals rather than a closed minimax recursion for all such classes. The physical theorem supplies a matched preparation segment in a continuous two-parameter family with overlapping stochastic reports; its converse is not a finite parameter-grid check.

The stationary autonomous result in Section 6 has an explicit formula, but its stationary lower bound must not be applied to a finite-horizon clocked machine. Its finite-time upper bound is purposely conservative and exposes the slowing caused by finite transition precision. An arbitrary compact-family presentation still requires additional effective information to construct a kernel net. A subdivided positive-basis degree bound must not be called a degree bound for a global SOS hierarchy.

The physical result is a genuine consumer of the experimental and continuation root. It does not make the independent A2 geometric proofs dependent on this root. The original B4 normalized-resolvent, range, compactness and control-transfer obligations, and the broad C2 strict-dual, form, rigidity and optional-projection obligations, are not proved by our finite-state theorem. Their statements, derivations and current status remain in the retained historical pipeline. No aggregate closure is inferred from preserving their source files.

Finally, the rational certificate (10), polynomial identity (12), and residual transitions are independently reproducible. The program in the accompanying source checks their exact identities and includes deliberately invalid controls. These checks are not substitutes for the proofs above. Neither compilation, preservation of earlier pages nor a successful finite test establishes novelty or an editorial decision.

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